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R. V. COLLEGE OF ENGINEERING

Autonomous Institution affiliated to VTU

IV Semester B. E. Fast Track Examinations July-18

Computer Science and Engineering

THEORY OF COMPUTATION

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

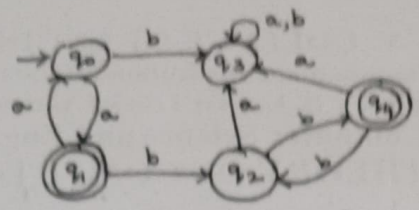
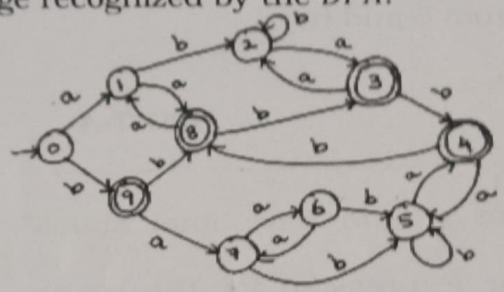
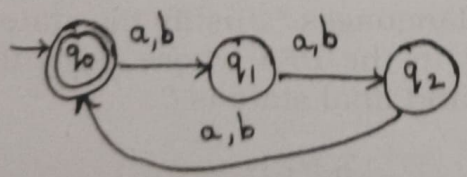
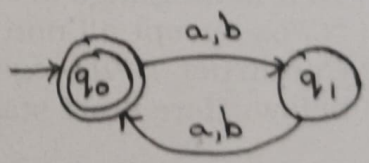
PART-A

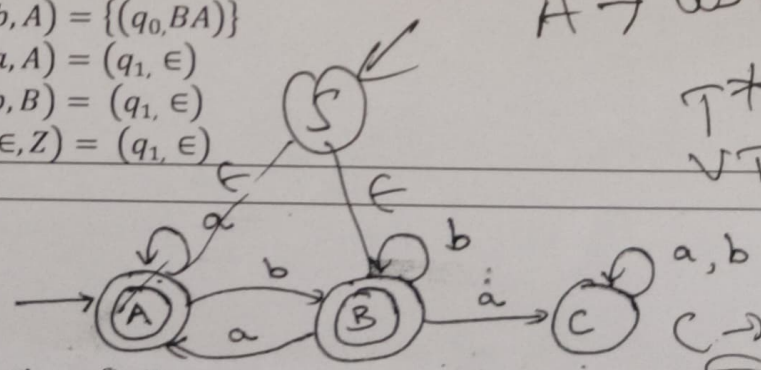
1	1.1	Define $\in$-NFA.	01
	1.2	"All automata should have final states". State true or false with justification.	02
	1.3	Write Regular Expression for the language consisting of the set of strings over alphabet $\{a, b, c\}$ containing atleast one 'a' and atleastone 'b'.	01
	1.4	Given two finite automata $M_1$and M_2, are there any strings that are accepted by neither?	02
	1.5	State Pumping Lemma for regular language.	01
	1.6	Construct two different Parse trees for the string "$abbabbab$" generated by the CFG, $S \rightarrow aS\|b\,S\|\,b$	02
	1.7	What is the language generated by the CFG with the productions: $S \rightarrow aSb\|aaSb\|\in$	01
	1.8	Define language accepted by PDA.	02
	1.9	"$CFGs$ accept all non-linear languages." Justify the statement given.	02
	1.10	Construct $DPDA$ equivalent to the DFA whose transition table is as below. Here start state is A and final state is C.	

	0	1
A	B	A
B	B	C
C	C	C

	1.11	Draw the Chomsky hierarchy diagram.	02
	1.12	Give example for blank tape non-halting problem in TM.	02

PART-B

2	a	Derive a regular expression for the *DFA* given below using state elimination method. And also mention the language accepted by the given automata. 	04
	b	For each of the regular expression below, find an *NFA*-ϵ recognizing the language corresponds to the regular expressions by applying the Kleen's theorem part-I. Do not attempt to simplify the answer. i) $(a^* + b^* + c^*)^*$ ii) $(ab + (bab)^*)(a)$	04
	c	Minimize the given *DFA* below, using table filling algorithm and identify the language recognized by the *DFA*. 	08
3	a	Using Pumping Lemma for regular sets show that the language below is not regular. $L = \{0^n \mid n \text{ is prime}\}$	04
	b	Let M_1 and M_2be the *DFAs* as shown below, accepting the languages L_1 and L_2 respectively. Draw *DFAs* accepting the following languages. i) $L_1 \cup L_2$ ii) $L_1 \cap L_2$ iii) $L_1 - L_2$ M_1:  M_2: 	06
	c	Given below a *CFG* G find a *CGF* G' in *GNF* generating $L(G) - \{\epsilon\}$. $S \rightarrow AB$ $A \rightarrow BS \mid a$ $B \rightarrow SA \mid b$	06
		OR	
4	a	Design *CFG* to generate the language i) $L = \{a^i b^j c^k : i + j = k,\ i \geq 0,\ j \geq 0\}$ ii) $L = \{a^n b^m c^n : n,\ m \geq 1\}$	04

	b	Convert the $CFG(V,T,P,S)$ with the following productions to CNF. $S \to abAB$ $A \to bAB \mid \in$ $B \to Baa \mid A \mid \in$	06
	c	Consider the Grammar $E \to +EE \mid *EE \mid -EE \mid x \mid y$. i) Find leftmost and rightmost derivations and also a derivation tree for the string $+ * - xyxy$. ii) Define sentential forms and also Unambiguous Grammar. And prove that the given Grammar is unambiguous.	06
5	a	Prove the CFL's are not closed under intersection using example.	04
	b	Define $DPDA$. Construct a $DPDA$ to accept the Language $L = \{x \in \{0,1\} \mid n_0(x) > n_1(x)\}$.	06
	c	State and prove pumping lemma for CFL.	06
		OR	
6	a	Construct a PDA to accept the Language $L = \{a^n b^{2n} \mid n \geq 1\}$. And trace the string $abb, aabb$.	04
	b	For the given grammar obtain the corresponding PDA with empty stack acceptance $S \to aABB \mid aA$ C $A \to aBB \mid a$ $B \to bBB \mid A$ $C \to a$	04
	c	Convert the given $PDA(M)$ into equivalent $CFG(G)$. $M = (\{q_0, q_1\}, \{a,b\}, \{A,B,Z\}, \delta, q_0, Z)$ where δ is defined as., $\delta(q_0, a, Z) = (q_0, AZ)$ $\delta(q_0, b, Z) = (q_0, BZ)$ $\delta(q_0, a, A) = \{(q_0, AA), (q_1, \in)\}$ $\delta(q_0, b, B) = \{(q_0, BB), (q_1, \in)\}$ $\delta(q_0, a, B) = \{(q_0, AB)\}$ $\delta(q_0, b, A) = \{(q_0, BA)\}$ $\delta(q_1, a, A) = (q_1, \in)$ $\delta(q_1, b, B) = (q_1, \in)$ $\delta(q_1, \in, Z) = (q_1, \in)$ $A \to abB$ $T^* V$ $V T^*$	08
7	a	 Convert the given finite automata to: 4 i) Right Linear Grammar. 4 ii) Left Linear Grammar. $C \to aC \mid bC$ $A \to aA \mid bB \mid \in$ $B \to aA \mid bB \mid \in \mid a$ 6	08
	b	Define Linear Bounded Automata with proper example.	04
	c	Define Context Sensitive Grammar (CSG). And design CSG for the language $L = \{a^n b^n c^n \mid n \geq 1\}$.	04

$\alpha \to B$ $|\alpha| \leq |B|$ $(V \cup T)^+$

8	a	Design a TM for the following language i) $L = \{ww^R \mid w \text{ is any string of } 0's \text{ and } 1's\}$ and also trace the string '0110" and '11100111".	06
	b	Design a TM to subtract 2 non-negative integers. Function is precisely defined below: $f = m - n \quad if\ m \geq n$ $= 0 \quad otherwise$	04
	c	i) Design unrestricted Grammar for the Language: $L = \{ww : w \in (a,b)^*\}$ X 3 ii) Solve the PCP given below:	

	List A	List B
i	w_i	x_i
1	10	101
2	011	11
3	101	011

3

06

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IV Semester B. E. Examinations April/May-16

Common to ISE / CSE

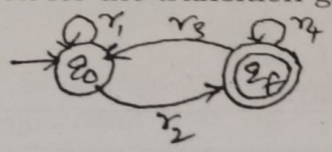
THEORY OF COMPUTATIONS

Time: 03 Hours *Maximum Marks: 100*

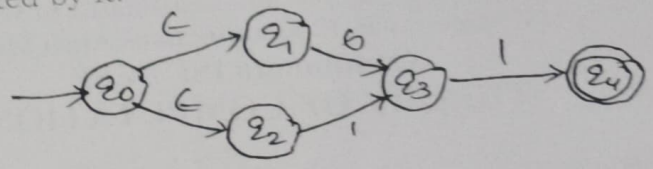
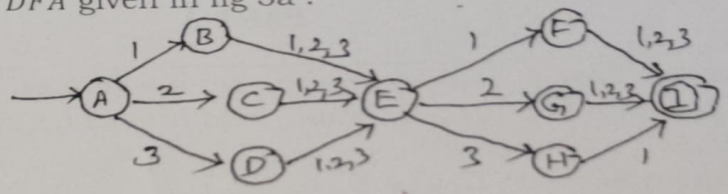
Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in the first three pages of the answer book only.
2. Answer FIVE full questions from Part B.

PART-A

1	1.1	For the language $L = \{ab, c\}$ over the set $A = \{a, b, c\}$, find L^3.	01
	1.2	Given $L_1 = \{a^i b^i c^j \ / \ \ i,j \geq 1\}$ and $L_2 = \{a^i b^j c^j \ / \ \ i,j \geq 1\}$, find $L_1 \cap L_2$.	01
	1.3	If $L = \{a, bba, aaa\}$ then $L^0 =$ _____.	01
	1.4	The relation between Σ^* and Σ^+ is _____.	01
	1.5	Write the primitive regular expressions.	01
	1.6	Write the regular expression for the language: $L = \{a^{2n} b^{2m} b : n, m \geq 0\}$ over $\Sigma = \{a, b\}$.	01
	1.7	Find the regular expression for the transition graph in Fig 1.7:  Fig 1.7	01
	1.8	Draw the plan for showing the equivalence of four different notations for regular languages.	01
	1.9	If $G = (V, T, P, S)$ is a CFG, then any string α in $(VUT)^*$ such that $S \overset{*}{\Rightarrow} \alpha$ is a _____.	01
	1.10	The transition function δ for the PDA is given by _____.	01
	1.11	_____ of a CFL and regular language is always CFL.	01
	1.12	CNF grammar has atmost _____ productions.	01
	1.13	Define instantaneous description (ID) of pushdown automata.	01
	1.14	A pushdown automaton is essentially _____ with a stack data structure.	01
	1.15	Draw the organization of constructions showing equivalence of three ways of defining the $CFLs$.	01
	1.16	Define deterministic pushdown automata.	01
	1.17	The languages accepted by TMs are called _____.	01
	1.18	Show the relationship between recursive RE and non-RE languages.	01
	1.19	The _____ symbol is in Γ' but not in Σ of TM.	01
	1.20	Every language accepted by a multiple TM is _____.	01

PART-B

2	a	Design a DFA that accepts those words from $\Sigma = \{a, b\}$ where the number of b's is divisible by three. Sketch the state table diagram of finite automaton M.	05
	b	Convert the given $\varepsilon - NFA$ in fig 2b to its equivalent DFA. Also define the language accepted by it.  Fig 2b	08
	c	Write the differences between NFA and DFA.	03
		OR	
3	a	Minimize the DFA given in fig 3a .  Fig 3a	08
	b	Design an NFA to accept strings from $\Sigma = \{a, b\}$ ending with ab or ba.	04
	c	Construct a ε-NFA that accepts decimal numbers consisting of i) an optional + or - sign; ii) a string of digits; iii) a decimal point; iv) another string.	04
4	a	Obtain regular expression for the following: i) over $\Sigma = \{a, b\}$ and $L(r) = \{a, bb, aa, abb, ba, bbb, \ldots\}$; ii) over $\Sigma = \{0,1\}$ and $L(r) = \{all\ strings\ containing\ substring\ 00\}$; iii) over $\Sigma = \{0,1\}$ and without two consecutive zeroes.	08
	b	Consider the DFA: $\begin{array}{c\|cc} \delta & 0 & 1 \\ \hline \rightarrow q_1 & q_2 & q_1 \\ & q_2 & q_3 \\ & q_3 & q_2 \end{array}$ i) Give all the regular expressions $R_{ij}^{(o)}$; ii) Obtain the regular expression by eliminating q_2.	08
		OR	
5	a	State and prove pumping lemma for regular languages.	08
	b	Prove that $L = \{0^n 1 0^n \ / \ n \geq 1\}$ is not regular.	04
	c	Prove that class of regular languages is closed under union and concatenation.	04

6	a	Obtain a CFG to generate the following language: $L = \{W/W\varepsilon(a+b)^{*}$ *and all strings of even length*$\}$.	05
	b	Obtain the LMD, RMD and parse tree for the string $aabbabab$ where the grammar G is given by $S \to aB/bA, A \to a/as/bAA,\ B \to b/bS/aBB$.	07
	c	Show that the language of all non-null strings of a's defined by CFG $G = \{\{S\},\{a\},\{S \to aS/Sa/a\},S\}$ is ambiguous.	04
		OR	
7	a	Eliminate all ϵ −productions from the grammar: $S \to BAAB, A \to 0A2/2A0/\epsilon,\ B \to AB/1B/\epsilon$.	03
	b	Remove useless symbols and production from $G = (V,T,S,P)$, $V = \{S,A,B,C\}$, $T = \{a,b\}$ with productions $S \to aS/A/C,\ A \to a, B \to aa,$ $C \to acb$.	03
	c	Remove all unit productions from: $S \to Aa/B, B \to A/bb, A \to a/bc/B$.	03
	d	Reduce the $CFG\ G$ into Chomsky normal form given by following productions: $S \to aAc, A \to aB/bAB, B \to b, C \to c$.	07
8	a	Design a PDA to accept $L = \{WCW^{R}\ /\ W \in \{a,b\}^{*}\}$. Show the sequence of IDs to accept the string $abcba$.	10
	b	Convert the grammar $S \to aSa/bSb/\lambda$ to a PDA that accepts the same language by empty stack.	06
		OR	
9	a	Construct a PDA to accept balanced parenthesis and trace the PDA for the string $(()())$.	08
	b	Convert the following PDA to Grammar: $P = \{\{q_0\},\{b,c\},\{z_0,z_1\},\delta,\varepsilon_0,z_0\}$.	08
10	a	Design a Turing machine to accept palindrome over $\{a,b\}$. Give the sequence of IDs for the strings "$abaa$" and "aa".	10
	b	Prove that if L is recursive language, then so is complementation of L.	06
		OR	
11	a	Define Post's correspondence problem (PCP). Solve the following instance of PCP:	

	A	B
i	w_i	x_i
1	10	101
2	01	100
3	0	10
4	100	0
5	1	010

			08
	b	Write short notes on: i) Multitape Turing machine; ii) Chomsky hierarchy.	08

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IV Semester B. E. Examinations April/May-18

Computer Science and Engineering

THEORY OF COMPUTATIONS

Time: 03 Hours *Maximum Marks: 100*

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B.In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6

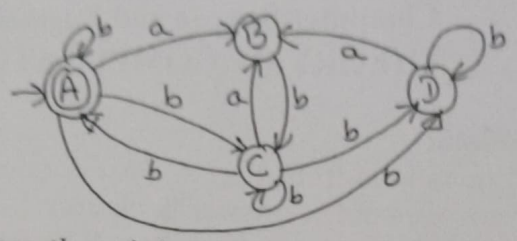
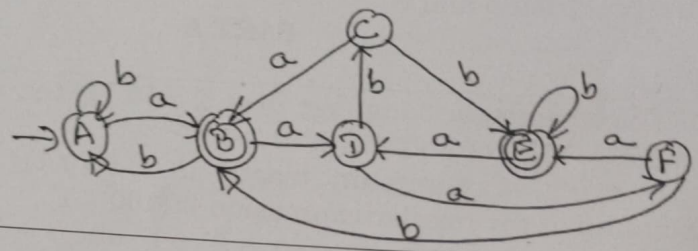
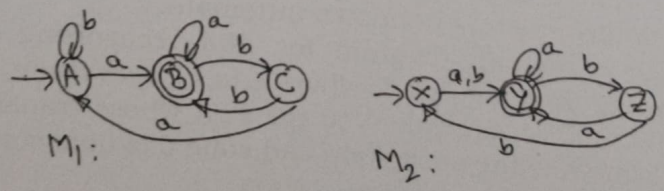
PART A

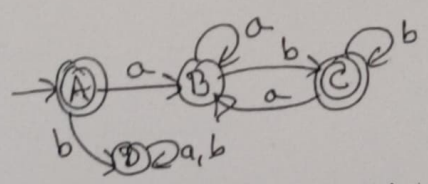
1	1.1	Give DFA accepting the language over $\Sigma = \{0,1\}$, the set of all strings that either begin or end (or both) with 01.	02
	1.2	Find the string of minimum length in $\Sigma = \{0,1\}$ not in the language corresponding to the regular expression $0^*(100^*)^*1^*$.	01
	1.3	Consider two regular expressions $r = 0^* + 1^*$ and $s = 01^* + 10^* + 1^*0 + (0^*1)^*$. Find a string corresponds to r but not to s.	01
	1.4	For the regular expression $((ab)^*b + ab^*)^*$ find an equivalent $NFA-\epsilon$ by applying the Klein's theorem part . Do not attempt to simplify the answer.	01
	1.5	What is the language generated by the CFG with the productions $S \rightarrow aSa \mid bSb \mid \epsilon$.	01
	1.6	Define leftmost derivation. Give leftmost derivation for the string abaaba in the grammar with productions: $S \rightarrow aSaSbS \mid aSbSaS \mid bSaSaS \mid \epsilon$	02
	1.7	Define deterministic pushdown automata.	01
	1.8	Give the transition diagram for PDA recognizing the language for PDA recognizing the language $L = \{w \mid w \in \{a,b\}^* \text{ such that } n_a(w) = n_b(w)\}$.	02
	1.9	Construct $DPDA$ equivalent to the PDA whose transition table is as below. There state A is the start state and state C is final state.	02

	0	1
A	B	A
B	B	C
C	B	A

	1.10	Say whether the following statement is $TRUE$ or $FALSE$. "Every regular language can be accepted by a $DPDA$ with only two states in which there are no ϵ – transitions and no symbols are ever removed from the stack.	01
	1.11	Identify useless variables in the CFG below: $S \rightarrow A \mid B \mid C, A \rightarrow aAa \mid B \mid a, B \rightarrow bb \mid bB, C \rightarrow aCaa \mid D, D \rightarrow baD \mid abD$.	01
	1.12	Recursively enumerable languages are also called as _____.	01
	1.13	Obtain Turing machine to accept the language $L = \{w \mid w \in \{a,b\}^* \text{ and } w \text{ ends with } 011\}$.	02
	1.14	Algorithm written for Turing machine that always halts is said to be _____.	01
	1.15	If L_1 and L_2 are recursively enumerable language over Σ then $L_1 \cap L_2$ is _____.	01

PART B

2	a	Define regular expressions. Give regular expression which generates the following languages over the alphabet $\Sigma = \{0, 1\}$. i) Strings that do not end with 01. ii) Strings that do not contain the substring 00.	06
	b	Consider the *NFA* below, using the subset construction method draw the *DFA* accepting the language which is same as the language accepted by *NFA*. 	
	c	For the *DFA* below, use the minimization algorithm to find a minimum state *DFA* recognizing the same language [By table filling approach]. 	06
3	a	State and prove pumping lemma for regular languages.	04
	b	Using pumping lemma show that the language $L = \{xy \mid x, y \in \{0,1\}^* \text{ and } y \text{ is either } x \text{ or } x^r\}$ is not regular.	05
	c	Let M_1 and M_2 are the *DFAs* as shown below accepting languages L_1 and L_2 respectively. Draw *DFAs* accepting the following languages: i) $L_1 \cap L_2$ ii) $L_1 - L_2$  M_1: M_2:	05
		OR	06
4	a	Define *CFG* and language generated by *CFG*. Construct *CFG* to generate the following languages: i) $L_1 = \{a^i b^j c^k \mid i = j + k\}$ ii) $L_2 = \{a^i b^j c^k \mid j = i \text{ or } j = k\}$	06
	b	Define ambiguity in *CFG*. Show that the *CFG* below is ambiguous: $S \rightarrow ABA$ $A \rightarrow aA / \epsilon$ $B \rightarrow bB / \epsilon$	04
	c	Given below a *CFG* G, find a *CFG* G' in *GNF* generating $L(G) - \{\epsilon\}$ $S \rightarrow AB \mid ABC$ $A \rightarrow BA \mid BC \mid a \mid \epsilon$ $B \rightarrow AC \mid CB \mid b \epsilon$ $C \rightarrow BC \mid AB \mid A \mid c$	06

5	a	Define PDA and Instantaneous description (ID). Construct PDA to accept set of all palindromes over $\{a,b\}$. Show by IDs the string $aabaabaa$ is accepted.	06
	b	List the steps to convert the given CFG to equivalent PDA by empty stack. Convert the CFG below to its equivalent PDA by empty stack. Show that the string $a+(a*b)$ is generated by the given grammar and the same is accepted by the equivalent PDA. $S \to S+S \mid S*S \mid S-S \mid (S) \mid a \mid b$	04
	c	Find the equivalent CFG for the PDA below $\delta(q_0,a,Z)=(q_0,AZ)$ $\delta(q_0,a,A)=(q_0,AA)$ $\delta(q_0,b,A)=(q_1,\epsilon)$ $\delta(q_1,b,A)=(q_1,\epsilon)$ $\delta(q_1,\epsilon,Z)=(q_0,Z_0)$ Show that the string $aaabbb$ is accepted by the given PDA and the same is generated by its equivalent grammar.	06
		OR	
6	a	State and prove pumping lemma for CFLs. Show that the language $L=\{a^ib^jc^k \mid i<j<k\}$ is not context free.	06
	b	Let $L_1=\{a^ib^jc^k \mid i \geq j \text{ or } i \geq k\}$ and $L_2=\{a^ib^jc^k \mid i \neq j \text{ or } i \neq k\}$. Show that L_1 and L_2 are context free but L_1-L_2 is not context free.	06
	c	Give an example of a $DPDA$ M accepting a language L for which the language accepted by the machine obtained from M by reversing accepting and non-accepting states is $\overline{L}$.	04
7	a	Describe the language generated by the regular grammar given below. Draw its equivalent NFA. $S \to aA \mid bC \mid b, A \to aS \mid bB$ $B \to aC \mid bA \mid a, C \to aB \mid bS$	04
	b	For the DFA accepting L shown below, find an equivalent left linear grammar generating $L-\{\epsilon\}$. Show that the string $aabab$ is generated by the grammar.  	04
	c	Define Linear Bounded Automata. Construct LBA to accept the language $L=\{a^nb^nc^n \mid n \geq 1\}$. Show that the string $aabbcc$ is accepted.	04
	d	Defend context sensitive grammar. Give context sensitive grammar to generate the language $L=\{ww \mid w \in \{a,b\}^*\}$. Show that the string $abaaba$ is generated.	04
8	a	Define Turing machine and the language of TM. Design TM to perform the string copy operation. The string is constructed over $\Sigma=\{a,b\}$. Trace the machine for string bab.	06
	b	Define unrestricted grammar. Give unrestricted grammar to generate the language $L=\{a^nb^nc^n \mid n \geq 1\}$. Show that the string $aabbcc$ is generated.	04

c	Define post corresponding problem. Solve the following instance of post correspondence problem	06

	List A	List B
i	w_1	x_i
1	10	101
2	01	100
3	0	10
4	100	0
5	1	010

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(Autonomous Institution affiliated to VTU)

IV Semester B. E. Fast Track Examinations July-19

Computer Science and Engineering

THEORY OF COMPUTATIONS

Time: 03 Hours *Maximum Marks: 100*

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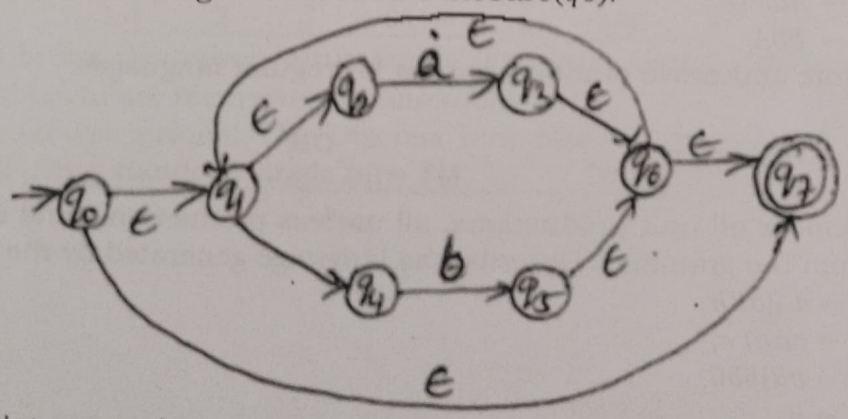
PART A

1	1.1	Consider two RE $r = 0*+1*$ and $s = 01*+10*+1*0+(0*1)*$ i. Find a string in $\{0,1\}^*$ corresponding to neither r nor s. ii. Find a string in $\{0,1\}^*$ corresponding to r and s	01
	1.2	Write ϵ–NFA for the Regular Expression $((0+1)(0+1))*$.	02
	1.3	Find a RE corresponding to subsets of $\{0,1\}*$, whose second symbol from left and right end is same.	01
	1.4	Define left recursion. Eliminate left recursion from the following grammar: $S \rightarrow (L)\|x$ $L \rightarrow L,S\|S$	02
	1.5	Transition Table of Turing machine is given below.	

State q	Input X	Move $\delta(q,X)$
$q0$	Δ	$(q1,\Delta,R)$
$q1$	A	$(q2,a,R)$
$q2$	B	$(q3,b,R)$
$q3$	A	$(q3,a,R)$
$q3$	Δ	$(q4,\Delta,S)$

Define the language accepted by Turing machine assuming $q0$ as initial sate and $q4$ as final sate. 01

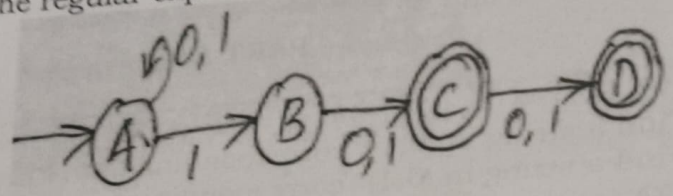
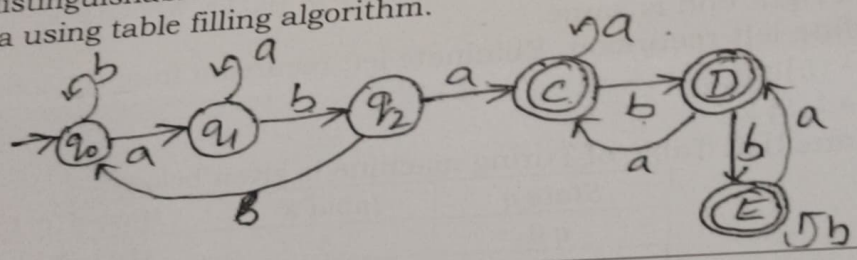
1.6 Consider the following ϵ–NFA. Find ϵ-closure($q0$).



	1.7	Differentiate recursive and recursively enumerable languages.	01
	1.8	For $\Sigma=\{0,1\}$, design a Turing machine that accepts the language denoted by the regular expression $(a+b)*ab(a+b)*$.	02

	1.9	Define Context Sensitive Grammar.	01
	1.10	CFLs are not closed under _____ and _____.	01
	1.11	Define inherently ambiguous grammar and give an example for inherently ambiguous language.	02
	1.12	Consider homomorphism h from alphabet $\{0,1,2\}$ to $\{a,b\}$ defined by $h(0) = ab, h(1) = b$ and $h(2) = aa$. Find $h(0210), h(2201), h(1*02*)$ and $h^{-1}(ababb)$.	02
	1.13	Show a derivation tree for the string aabbbb with the grammar and describe the language generated by this grammar $S \rightarrow AB \mid \epsilon$ $A \rightarrow aB,$ $B \rightarrow Sb.$	02

PART B

2	a	Discuss the applications of Regular Expressions with an example.	04
	b	Construct the regular expression from following automata:  	06
	c	Define distinguishable and indistinguishable states. Minimize the following automata using table filling algorithm.  	06

3	a	Define CFG. Construct a CFG for the following languages: i. $L = \{a^n b^m : n <= m + 3\}$ ii. $L = \{a^n W W^R b^n : W \in (a,b)*\}$	05
	b	Define CNF. Convert the following grammar into Chomsky Normal form: $S \rightarrow AB \mid aB$ $A \rightarrow aab \mid \epsilon$ $B \rightarrow bbA$	06
	c	State and prove pumping lemma for regular languages.	05
		OR	
4	a	Remove all unit-productions, all useless productions, and all ϵ-productions from the grammar. Describe the language generated by the grammar. $S \rightarrow aA \mid aBB,$ $A \rightarrow aaA / \epsilon,$ $B \rightarrow bB \mid bbC,$ $C \rightarrow B.$	05

	b	Define LMD,RMD and Ambiguous grammar. Show that the following grammar is ambiguous. Construct An unambiguous grammar equivalent to this grammar. $S \rightarrow AB \mid aaB,$ $A \rightarrow a \mid Aa,$ $B \rightarrow b.$	07
	c	Discuss the applications of Context Free Grammars	04
5	a	Discuss the languages accepted by PDA with an example for each	05
	b	Construct PDA that accept the language $L = \{N_a(w) = 2N_b(w)\}$ on $\Sigma = \{a, b\}$	07
	c	Construct the NPDA corresponding to the grammar $S \rightarrow aABB \mid aAA,$ $A \rightarrow aBB \mid a,$ $B \rightarrow bBB \mid A.$	04
		OR	
6	a	Consider the NPDA with transitions $\delta(q_0, a, Z) = \{(q0, AZ)\},$ $\delta(q_0, a, A) = \{(q0, A)\},$ $\delta(q_0, b, A) = \{(q1, \in)\},$ $\delta(q_1, \in, Z) = \{(q2, \in)\},$ using q0 as initial state and q2 as final state. Construct an equivalent CFG.	07
	b	Prove that CFLs are closed under Union, Concatenation and Closure.	05
	c	Show that the language L={WW | W ∈ (0,1)*} is not context free.	04
7	a	Define Linear Bounded Automata. Design a linear bounded automata to accept the language $L = \{a^n b^n c^n \mid n >= 0\}$.	08
	b	Define left linear and right linear grammars. Construct right and left-linear grammar for the language $L = \{a^n b^m : n >= 2, m >= 3\}$. Construct DFA for both left and right linear grammars.	08
8	a	Define PCP. Given an instance of PCP, check whether this instance has a solution.	04

	List A	List B
i	Wi	Xi
1	110	110110
2	0011	00
3	0110	110

	b	If L_1 and L_2 are recursively enumerable languages over Σ, then prove that $L_1 \cap L_2$ and $L_1 \cup L_2$ are recursively enumerable.	06
	c	Define multi-dimensional TM. Discuss how Multi-dimensional TM can be simulated using standard single tape TM.	06